

Roll No.

TEC-501

**B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

DIGITAL SIGNAL PROCESSING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find the all possible ROCs and inverse Z transform of the following : (CO1)

$$H(z) = (3 - 4z^{-1}) / (1 - 3.5z^{-1} + 1.5z^{-2})$$

- (b) Explain the basic elements Digital Signal Processing system. Point the advantage of digital over analog signal processing. (CO1)

- (c) Explain the following : (CO1)

(i) Cross Correlation and Auto-Correlation

(ii) Linear Convolution

P. T. O.

2. (a) Prove the 'DIFFFT' algorithm for computation on $N = 8$ point DFT.

(CO2)

- (b) Prove that the multiplication of two DFTs is equivalent to circular convolution.

(CO2)

- (c) Find four point DFT of the sequence $x(n) = \{0, 1, 2, 3\}$.

(CO2)

↑

Compare the computational complexity of DFT and FFT algorithm.

(CO2)

3. (a) Prove that Direct Form I Direct Form II structure of the IIR digital filter.

(CO2)

- (b) Determine the Direct Form II and cascade realization of the following system :

(CO3)

$$y(n) = \left(\frac{3}{4}\right)y(n-1) - (1/8)y(n-2) + x(n) - (1/3)x(n-1)$$

- (c) Prove and draw the structure for linear phase FIR Filter for M even.

Obtain the Linear phase structure of the following system :

(CO3)

$$H(z) = (1 + 0.5z^{-1} + z^{-2})(1 + 0.5z^{-1} + z^{-2})$$

4. (a) Prove the 'Impulse Invariant Technique' for design of IIR Filter. (CO4)

- (b) Design a single pole low pass digital filter with a 3dB bandwidth of 0.2π , using the Bilinear transformation applied to the analog filter :

(CO4)

$$H(s) = \frac{\Omega}{(s + \Omega)}$$

where Ω is the 3dB bandwidth of analog filter.

- (c) Explain the design steps of Butterworth filter. (CO4)

(3)

5. (a) Explain Hamming, Hanning and Blackman, Rectangle, Bartlett and Kaiser windows for design of FIR filter. (CO5 and CO6)

- (b) The desired response of low-pass filter is :

$$H_d(e^{j\omega}) = e^{-j3\omega}; -3\pi/4 < \omega < 3\pi/4$$

$$= 0; 3\pi/4 < \omega < \pi$$

Determine $H(e^{j\omega})$ for $m = 7$, using Hamming Window.

- (c) What is 'Gibbs phenomenon' ? Why this phenomenon is developed in rectangular window and it is eliminated or reduced ? Discuss the windowing method of FIR filter design.

Roll No.

TEC-502

**B. TECH. (ECE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

COMMUNICATION SYSTEMS-II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) Find the Nyquist sampling rate and Nyquist interval for each of the following signals :

(1) $\sin c(100\pi t)$

(2) $\sin c^2(100\pi t)$

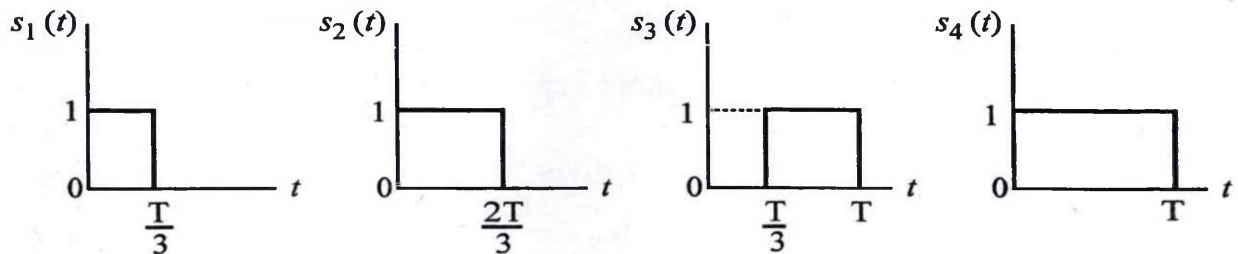
(ii) The spectral range of bandpass signal extends from 10 MHz to 10.4 MHz. Find the minimum sampling rate. (CO1)

(b) (i) Describe the midtread and midriser type characteristics of uniform quantizer with a suitable diagram. (CO1)

P. T. O.

- (ii) A binary channel with bit rate is equal to 36,000 bps is available for PCM voice transmission. Evaluate the appropriate values of the sampling rate f_s , the quantization levels, and the number of binary digits. (CO1)
- (c) (i) Determine the output SQNR in a DM system for 1 kHz sinusoidal, sampled at 32 kHz without slope overload distortion and followed by 4 kHz post-reconstruction filter. (CO1)
- (ii) Determine the minimum step size required to avoid slope overload distortion for a Delta-modulator operating at 32,000 sample/sec to track the signal : (CO1)
- $$x(t) = 125t[u(t) - u(t-1)] + [250 - 125t][u(t-1) - u(t-2)],$$
- where $u(t)$ is the unit step function.
2. (a) Determine the power spectral density for the Manchester format. (CO2)
- (b) What is Nyquist criterion of zero ISI ? Explain. (CO2)
- (c) In a baseband communication link, frequencies upto 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter symbol interference, determine the maximum possible signaling rate in symbols per sec. (CO2)
3. (a) Prove that the spectrum of the output signal of a matched filter with the matched signal as input is, except for a time delay factor, proportional to the energy spectral density of the input signal. (CO1 and CO3)

- (b) Figure shown below displays the waveforms of four signals $s_1(t)$, $s_2(t)$, $s_3(t)$ and $s_4(t)$. Using the Gram-Schmidt orthogonalization procedure, find an orthonormal basis for this set of signals. (CO1 and CO3)



- (c) (i) Let $g(t) = e^{-\pi/t^2}$, and $h(t)$ is a filter matched to $g(t)$. If $g(t)$ is applied as input to $h(t)$, find the Fourier transform of the output. (CO1 and CO3)
- (ii) What is the slope overload distortion and granular noise in Delta modulation and how it is removed in Adaptive Delta modulation? (CO1 and CO3)
4. (a) Derive the power spectral density of BPSK and BASK signals and draw the power spectrum of each. (CO4)
- (b) Prove that the reconstruction of the DPSK signal is independent of the choice of the extra bit. (CO4)
- (c) Represent orthogonal BFSK signals in the signal space and find the Euclidean distance. (CO4)
5. (a) A DMS X has six symbols x_1, x_2, x_3, x_4, x_5 and x_6 with $P(x_1) = 0.30$, $P(x_2) = 0.25$, $P(x_3) = 0.20$, $P(x_4) = 0.12$, $P(x_5) = 0.08$, and $P(x_6) = 0.05$.
- (i) Construct a Shannon-Fano code for x , and calculate the efficiency of the code. (CO5 and CO6)
- (ii) Repeat for the Huffman code and compare the results. (CO5 and CO6)

- (b) (i) A communication channel with AWGN operating at a signal at a signal to noise ratio $\text{SNR} \gg 1$ and bandwidth B has capacity C_1 . If the SNR is doubled keeping B constant. Determine the resulting capacity C_2 . (CO5 and CO6)
- (ii) Show that the channel capacity of an ideal AWGN channel with infinite bandwidth is given by $C_\infty = 1.443 \frac{S}{\eta}$ bps, where S is the average signal power and $\frac{\eta}{2}$ is the power spectral density of white Gaussian noise. (CO5 and CO6)
- (c) Prove that the upper bound on the value of entropy $H(X)$ of a source X is $\log_2 M$, where M is the size of the alphabet of X . (CO5 and CO6)

Roll No.

TEC-503

B. TECH. (ECE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
MICROCONTROLLER AND EMBEDDED SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Classify the microcontrollers and explain the features of 8051 microcontroller. (CO1)
- (b) The following shows crystal frequency for following different 8051-based system. Find the period of the machine cycle in each case : (CO1)
 - (i) 11.0592 MHz
 - (ii) 18 MHz
 - (iii) 20 MHz
 - (iv) 22 MHz

P. T. O.

- (c) Differentiate between RISC and CISC Architecture, Write the code to push R5 and A onto the stack and then pop them back them into R2 and B, where $B = A$ and $R2 = R5$. (CO1)
2. (a) A switch is connected to pin P1.7. Write a program to check the status of the switch and make the following decision : (CO2)
- (i) If $SW = 0$, send "0" to P2
 - (ii) If $SW = 1$, send "1" to P2
- (b) Find out to which by each of the following bits belongs. Give the address of the RAM byte in hex : (CO2)
- (i) SETB 42H
 - (ii) CLR 67H
 - (iii) CLR 0FH
 - (iv) SETB 28H
 - (v) CLR 12
 - (vi) SETB 05
- (c) Explain the following pins : (CO2)
- (i) EA/VPP
 - (ii) PSEN
 - (iii) (RXD) P3.0 (TXD) P3.1
 - (iv) (INT0) P3.2 (INT1) P3.3
 - (v) (T0) P3.4 (T1) P3.5

(3)

3. (a) Assuming XTAL = 22 MHz, write a program to generate a square wave of frequency 1 kHz on P1.2. (CO3)
- (b) Assuming XTAL = 22 MHz, write a program to generate a pulse train of 2 seconds period on pin P2.4 Use Timer1 in mode 1. (CO3)
- (c) Explain the interrupts of 8051 microcontroller. Two switches are connected to pins P3.2 and P3.3. When a switch is pressed, the corresponding LEDs glow. Write a program to (i) light all LEDs connected to Port 0, if the first switch is pressed, (ii) light all LEDs connected to Port 2, if the second switch is pressed. (CO3)
4. (a) Explain the important features of ATMEGA328P. Also explain the utility of Ports. (CO4)
- (b) Explain the sketch of Arduino Software and basic digital and analog commands used. (CO4)
- (c) Write a program where according to the push button value at pin 7 the LED at Pin 13 works by using Arduino and also display the value of LED serially. (CO4)
5. (a) Write a program to interface LCD for sending command and data. (CO5, CO6)
- (b) Explain the concept of interfacing of ADC 0804 chip with timing diagram. (CO5, CO6)
- (c) Explain Serial communication interfacing with 8051 microcontroller. (CO5, CO6)

Roll No.

TEC-504

**B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ANTENNA AND WAVE PROPAGATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the significance of radiation pattern of an antenna and draw the radiation pattern of directional, isotropic and omnidirectional antenna ? What is the difference between antenna directivity and gain ? Explain antenna efficiency. (CO1)
- (b) Determine the magnetic field in far field region of Hertz dipole. Clearly mention the assumption made in evaluation of magnetic field. (CO1)
- (c) Determine the directivity for given radiation intensity :

$$U = \sin^2 \theta \sin^3 \phi \text{ (W/unit solid angle)}$$

$$0 \leq \theta \leq \pi, 0 \leq \phi \leq \pi$$

If the radiation resistance is 73 ohm and loss resistance is 50 ohm, determine the gain of the antenna. (CO1)

P. T. O.

2. (a) What are the different feeding techniques of parabolic reflector ? Explain with proper diagram. Determine the directivity if the diameter of parabolic antenna is 1 m, aperture efficiency is 70% and operating at 1 GHz. (CO2, CO6)
- (b) Discuss the Helical antenna in normal and axial mode. (CO2, CO6)
- (c) The output power of a 900 MHz mobile phone base station transmitter is 100 W. It is connected to an antenna having a gain of 15 dBi by a cable that has loss of 3 dB. Receiver antenna is kept at a distance of 25 km and gain of the receiver antenna is -1 dB. Calculate the power delivered to the receiver (i) if the system is impedance matched as well as polarization matched (ii) if the transmit antenna is circularly polarized and receiver antenna is linearly polarized in y-direction. (CO2, CO6)

3. (a) Show that the array factor for Uniform equally spaced linear array with progressive phase shift is given as $AF = \frac{\sin\left(\frac{N\psi}{2}\right)}{\left(\frac{N\psi}{2}\right)}$ where symbols have their usual meaning. (CO3)

- (b) Design an ordinary end-fire uniform linear array with only one maximum so that its directivity is 20 dB (above isotropic). The spacing between the elements is $\frac{\lambda}{4}$. Determine number of elements, overall length of the array (in wavelengths), angle where nulls occur and approximate half power beam width (in degree). Also draw the radiation

pattern of this array. If this array uses a half wavelength dipole as array element, what should be the far zone electric field ? (CO3)

- (c) Show that the Array Factor (AF) of six elements broadside linear array having non-uniform amplitude and equal spacing between the consecutive elements is given as : (CO3)

$$AF = a_1 \cos\left(\frac{kd}{2} \cos \theta\right) + a_2 \cos\left(\frac{3kd}{2} \cos \theta\right) + a_3 \cos\left(\frac{5kd}{2} \cos \theta\right)$$

where symbols have their usual meaning.

4. (a) Show that the radiation resistance of small circular loop antenna is

$$20\pi^2 \left(\frac{c}{\lambda}\right)^4, \text{ if the electric field of small loop antenna is given as :}$$

$$E_{\phi} = \frac{\eta k^2 a^2 I_0 e^{-jkr}}{4r} \sin \theta$$

where symbols have their usual meaning. (CO4, CO6)

- (b) Write the short notes on Yagi-Uda antenna and Log-periodic antenna. Design a Yagi-Uda antenna to be operated at central frequency of 200 MHz and expected directivity is 12 dB. This antenna will be fed through a coaxial cable. (CO4, CO6)

- (c) Design a rectangular microstrip antenna with dimension W and L, over a single substrate, whose centre frequency is 10 GHz. The dielectric constant of the substrate is 10.2 and the height of the substrate is 0.127 cm. Determine the physical dimensions W and L of the patch taking account field fringing. (CO4, CO6)

5. (a) Explain sub-refraction, super refraction and duct propagation for tropospheric wave propagation. Also explain optical horizon and radio horizon. (CO5, CO6)
- (b) Calculate the received power in a communication link having a path loss of 160 dB, transmit antenna gain of 20 dB, receive antenna gain of 25 dB and transmit power of 100 mW. If the minimum received power required to establish a reliable communication link is -80 dBm, what should be transmit power? (CO5, CO6)
- (c) What do you understand by critical frequency and maximum usable frequency? Establish the relationship between them. A high frequency radio link has to be established between two points on the earth 2000 km away. If the reflection region of the ionosphere is at a height of 200 km and has a critical frequency of 6 MHz, calculate the MUF for given path. (CO5, CO6)

Roll No.

TEC-552

**B. TECH. (ECE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

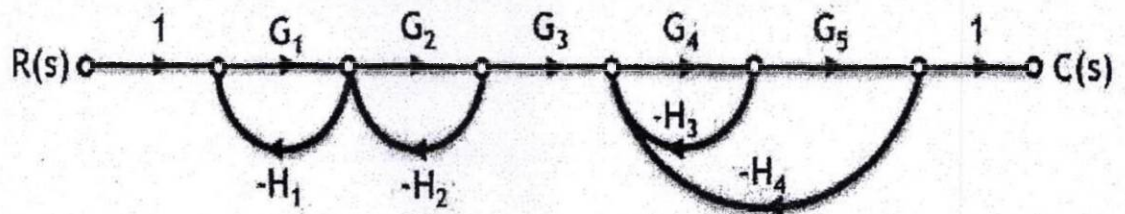
CONTROL SYSTEMS

Time : Three Hours

Maximum Marks : 100

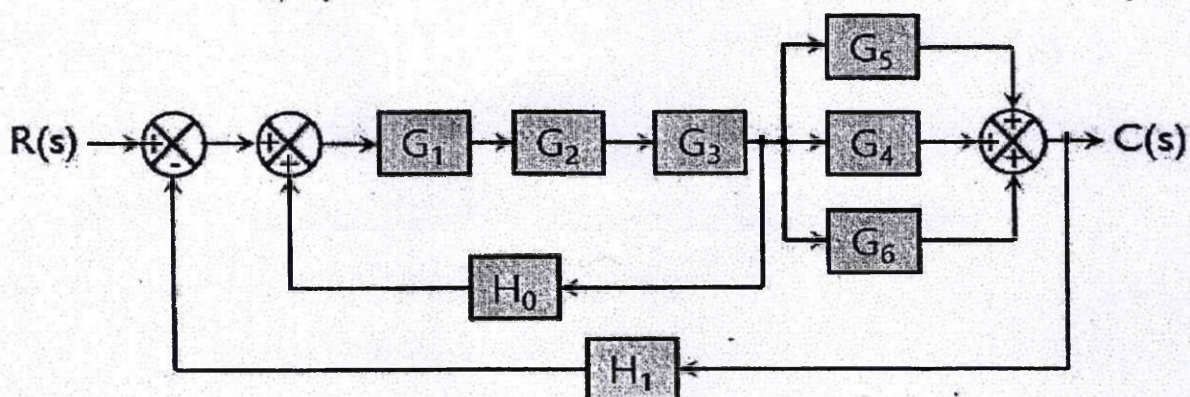
- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Draw and explain various components in block diagram of a feedback control system. Derive an expression for transfer function of a closed loop control system. (CO1)
- (b) Draw signal flow graph for the following set of simultaneous equations and hence derive the transfer function using Mason's gain formula ;

(CO1)



P. T. O.

- (c) Determine the overall transfer function $C(s)/R(s)$ of the block diagram shown in fig. : (CO1)



2. (a) Derive the expressions for Rise time and Settling time for a second order system. (CO2)
- (b) Derive unit step response for a second order unity feedback control system. Also represent it graphically and show important characteristics on the response. (CO2)
- (c) Analyse the effect of incorporating a PD controller to the second order closed loop control system. (CO2)
3. (a) Draw the polar plot for a system with unity feedback control system for $G(s) = 3/(s+1)$. (CO3)
- (b) Discuss various steps to draw a Root Locus Plot for given open loop transfer function. (CO3)
- (c) Draw Bode plot for a unity feedback control system with two open loop poles at origin. (CO3)

(3)

4. (a) Draw an electrical RC phase lead network and derive its transfer function. (CO3, CO4)
- (b) Check the stability using Routh's stability criteria for the system with the following characteristic equation : (CO3, CO4)

$$Q(s) = s^3 + 7s^2 + 25s + 39 = 0$$

- (c) For a unity feedback control system : (CO3, CO4)

$$G(s) = K / \{s(s+3)\};$$

Sketch Root Locus showing all important details on this plot. Comment on stability of the system.

5. (a) Prove any *five* properties of a State Transition Matrix. (CO5, CO6)
- (b) Derive transfer function model from state space model. (CO5, CO6)
- (c) Derive state space model for the system represented by the following differential equation : (CO5, CO6)

$$d^3y(t)/dt^3 + 4d^2y(t)/dt^2 + 5dy(t)/dt + 6y(t) = 7u(t)$$

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TEC-551

**B. TECH. (ECE-ESR) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

SENSOR TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain Saturation, Repeatability, Resolution, and Reliability of sensing devices in brief. (CO1)
- (b) Explain nonlinearity with the help of nonlinear devices. What do you understand by uncertainty in the sensing technology ? (CO1)
- (c) Explain Span (Full-scale input), Full-scale output. (CO1)
2. (a) Derive expression for total electric force exerted on a point positive unit charge by the charges ($Q_0, Q_1, \dots, Q_n$) situated at distances ($r_0, r_1, \dots, r_n$) respectively from the charges. (CO2)

P. T. O.

(2)

- (b) The total number of electrons in two identical bodies is typically in the order of 10^{20} each. Suppose due to some reason both bodies lost 5% of this number of electrons. Calculate the electrostatic force them separated at a distance of 1 m. Compare this with weight. Assume mass of each body is 50 kg and use point charge approximation. (CO2)
- (c) Explain in brief about Hall effect, Thermoelectric effect and their applications in sensors. (CO2)
3. (a) Derive expressions for gain in inverting and non-inverting operational amplifier. Also explain the applications of amplifiers. (CO3)
- (b) With the help of suitable diagram and example explain Analog to digital converters. (CO3)
- (c) Give brief notes about Ground planes, Seebeck noise. What do you understand by bypass capacitor ? (CO3)
4. (a) What is Potentiometric sensor and their various applications. (CO4)
- (b) What are the differences between Resistive and Capacitive sensors ? (CO4)
- (c) Write short notes on LVDT and RVDT. (CO4)
5. (a) Briefly describe the CMOS-Imaging sensor. (CO5/CO6)
- (b) What do you understand by Gas flame detectors, and humidity sensor. (CO5/CO6)
- (c) Why is sensing technology having important role for fabrication of sensing device ? (CO5/CO6)

Roll No.

TEC-591

B. TECH. (ECE) (IOT) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

TRANSDUCERS, ACTUATORS AND DISPLAY DEVICES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain thermoelectric effect. If cold junction of a thermocouple is 3°C , neutral temperature is 210°C , find inversion temperature. (CO1)
- (b) A piezoelectric crystal having dimension of $4\text{ mm} \times 4\text{ mm} \times 1.5\text{ mm}$ and voltage sensitivity of 0.045 V-m/N is used for force measurement. Calculate the force if the voltage developed is 120 V . (CO1)
- (c) Describe various static characteristics of sensors. (CO1)

P. T. O.

(2)

2. (a) A common base phototransistor has input impedance 25 ohm and output resistance of 120 kilohm. If a signal of 220 mV is applied between emitter and base, find the voltage amplification. Assume $\alpha = 1$. (CO2)
- (b) Explain working of interferometric optical transducer. (CO2)
- (c) Explain any *one* solid state transducer in detail. (CO2)
3. (a) Explain magnetostrictive transducer. How is it different from electrostrictive transducer ? (CO3)
- (b) What is microsensing for MEMS. Differentiate between capacitive sensing and piezoelectric sensing. (CO3)
- (c) What is the working principle of electromechanical transducers ? (CO3)
4. (a) Explain construction of solar cell with diagram and its I-V characteristics. (CO4)
- (b) What is the efficiency and spectral response of a solar cell ? (CO4)
- (c) Describe heterojunction solar cell. (CO4)
5. (a) What is plasma display ? Explain electroluminescence with diagram. (CO5, CO6)
- (b) What are the drawbacks of CRT ? Differentiate between LED and LCD. (CO5, CO6)
- (c) Find out the power generated by a solar cell of 6 m^2 area. If the input radiation is 850 W/m^2 and the efficiency of solar cell is 25%.

(CO5, CO6)

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TEC-503

B. TECH. (ECE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
MICROCONTROLLER AND EMBEDDED SYSTEMS

Time : Three Hours

Maximum Marks : 100

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(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

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(iv) Each sub-question carries 10 marks.

1. (a) Classify the microcontrollers and explain the features of 8051 microcontroller. (CO1)
- (b) The following shows crystal frequency for following different 8051-based system. Find the period of the machine cycle in each case : (CO1)
 - (i) 11.0592 MHz
 - (ii) 18 MHz
 - (iii) 20 MHz
 - (iv) 22 MHz

P. T. O.

- (c) Differentiate between RISC and CISC Architecture, Write the code to push R5 and A onto the stack and then pop them back them into R2 and B, where $B = A$ and $R2 = R5$. (CO1)
2. (a) A switch is connected to pin P1.7. Write a program to check the status of the switch and make the following decision : (CO2)
- (i) If $SW = 0$, send "0" to P2
 - (ii) If $SW = 1$, send "1" to P2
- (b) Find out to which by each of the following bits belongs. Give the address of the RAM byte in hex : (CO2)
- (i) SETB 42H
 - (ii) CLR 67H
 - (iii) CLR 0FH
 - (iv) SETB 28H
 - (v) CLR 12
 - (vi) SETB 05
- (c) Explain the following pins : (CO2)
- (i) EA/VPP
 - (ii) PSEN
 - (iii) (RXD) P3.0 (TXD) P3.1
 - (iv) (INT0) P3.2 (INT1) P3.3
 - (v) (T0) P3.4 (T1) P3.5

(3)

3. (a) Assuming XTAL = 22 MHz, write a program to generate a square wave of frequency 1 kHz on P1.2. (CO3)
- (b) Assuming XTAL = 22 MHz, write a program to generate a pulse train of 2 seconds period on pin P2.4 Use Timer1 in mode 1. (CO3)
- (c) Explain the interrupts of 8051 microcontroller. Two switches are connected to pins P3.2 and P3.3. When a switch is pressed, the corresponding LEDs glow. Write a program to (i) light all LEDs connected to Port 0, if the first switch is pressed, (ii) light all LEDs connected to Port 2, if the second switch is pressed. (CO3)
4. (a) Explain the important features of ATMEGA328P. Also explain the utility of Ports. (CO4)
- (b) Explain the sketch of Arduino Software and basic digital and analog commands used. (CO4)
- (c) Write a program where according to the push button value at pin 7 the LED at Pin 13 works by using Arduino and also display the value of LED serially. (CO4)
5. (a) Write a program to interface LCD for sending command and data. (CO5, CO6)
- (b) Explain the concept of interfacing of ADC 0804 chip with timing diagram. (CO5, CO6)
- (c) Explain Serial communication interfacing with 8051 microcontroller. (CO5, CO6)

BHPS-101-22 BAPSD ET 101 Understanding Political Theory

22 BANGSET 108 — Yoga 1

MBIT-102 — Fundamental of comp. Programming & C.

MBIT-103 — Biochemistry (MSc-120)

MBIT-105 — Cell & Developmental Biology (MSc110)

TMA — 101 — Engg. Mathematics

TEV — 101 — EVS.

THU — 101 — PC.

THU — 101 — Health & Fitness

THP — 101 — Health & Fitness

EVSC (Msc.) 001 — Fundamentals of Environmental Science.

MSc-160 — PC.

MSCFT-105 PC

BSC FT — Total

BSEM — 107 Healthy Beauty & Fitness

MBI-107 — P.C.

TMI — ERP.

BFSC-106b Elementary Forensic Science.

BFSL-107 — PC.

III Sem.

EVSC-013 → Environmental Economics,

EVSC-014 → Remote Sensing & Environmental Statistics

22 BSFTVA — Principles of Cookery
309

TMA310 — Advanced Engineering Mathematics (ECT) (TEC)

TRA-301 — Microprocessor & Microcontroller.